

AP CSP | Lesson 15: Randomness & Simulation

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B includes both integer endpoints. A and D omit endpoints; C includes nonintegers that RANDOM does not return.

Q2. Answer: D

D: values 1 and 2 give $2/6 = 1/3$. A assumes two categories are equal; B counts only one value; C counts the four failures.

Q3. Answer: A

A: one draw is stored and reused. B and C invent another draw or update. D chooses one possible result without justification.

Q4. Answer: C

C is $9/20$. A confuses an observation with the model; B treats ordinary variability as proof of error; D assumes past results force a future result.

Q5. Answer: A and D

A includes the upper endpoint. D is a possible repeat. B invents sampling without replacement; C incorrectly excludes matching draws. Both A and D are required.

Q6. Answer: B

B preserves the assumptions. A and D overclaim accuracy or absence of bias; C invents a model change that extra trials do not make.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: 3, 4, 5, 6, 7; five outcomes; $P(7) = 1/5$; $P(\text{even}) = 2/5$ because 4 and 6 qualify.

1 mark: Lists exactly 3, 4, 5, 6, 7.

1 mark: Counts five equally likely integer outcomes.

1 mark: Gives probability $1/5$ for 7.

1 mark: Gives $2/5$ and identifies 4 and 6.

Q8. 4 marks

Answer: Use $r \leq 3$: three of ten codes give 30%. Otherwise report No delay. $r < 3$ selects only 1 and 2, or 20%.

1 mark: Gives a condition selecting exactly three values, such as $r \leq 3$.

1 mark: Justifies probability $3/10 = 30\%$.

1 mark: Maps the remaining seven values to No delay.

1 mark: Explains that $r < 3$ gives $2/10 = 20\%$.

Q9. 4 marks

Answer: Counts: 1, 1, 2, 3. Final 3; fraction $3/4 = 0.75$. A fresh run need not match.

1 mark: Gives trace 1, 1, 2, 3.

1 mark: Gives final heads = 3.

1 mark: Gives observed fraction $3/4$ or 0.75.

1 mark: Explains that new random draws may yield a different result.

Q10. 4 marks

Answer: (a) $1/2$. (b) 0.53. (c) No; finite random counts vary. (d) No; the next trial still has probability $1/2$.

1 mark: Gives theoretical probability $1/2$.

1 mark: Calculates observed frequency $53/100 = 0.53$.

1 mark: Rejects guaranteed 500 Heads and cites finite variability.

1 mark: Rejects required correction because the next trial retains probability $1/2$.

Q11. 4 marks

Answer: First uses two draws; second one. Example odd sum: $3 = 1 + 2$. Doubling an integer yields only even sums. The first represents two independent fair dice.

1 mark: Identifies two draws versus one.

1 mark: Gives a possible odd sum with a valid pair.

1 mark: Explains that $2 \cdot r$ allows only even sums.

1 mark: Selects the first model for independent dice.

Q12. 4 marks

Answer: It can explore delay frequency cheaply. It assumes constant independent arrival outcomes. Lunch crowding may increase real delays. More trials still apply the same assumptions.

1 mark: Gives a relevant purpose, such as exploring delay rates or comparing assumptions.

1 mark: Identifies constant chance or independence as an assumption.

1 mark: Links an omitted factor to a possible effect on results.

1 mark: Explains that repetition does not add the omitted factor to the model.